

SQL Interview FAQ – Query Practice Sheet

1. Find the highest salary in each department

```
SELECT Dept,
       MAX(Salary) AS HighestSalary
FROM Emp
GROUP BY Dept;
```

2. Find the employee(s) with the highest salary in each department (Window Function)

```
SELECT Name,
       Dept,
       Salary
FROM
(
    SELECT Name,
           Dept,
           Salary,
           DENSE_RANK() OVER
           (
               PARTITION BY Dept
               ORDER BY Salary DESC
           ) AS rnk
    FROM Emp
) t
WHERE rnk = 1;
```

3. Find the employee(s) with the highest salary in each department (Correlated Subquery)

```
SELECT Name,
       Dept,
       Salary
FROM Emp e
WHERE Salary =
(
    SELECT MAX(Salary)
    FROM Emp
```

```
WHERE Dept = e.Dept  
);
```

4. Find the second highest salary

```
SELECT MAX(Salary)  
FROM Emp  
WHERE Salary <  
(  
    SELECT MAX(Salary)  
    FROM Emp  
);
```

5. Find the third highest salary

```
SELECT MAX(Salary)  
FROM Emp  
WHERE Salary <  
(  
    SELECT MAX(Salary)  
    FROM Emp  
    WHERE Salary <  
    (  
        SELECT MAX(Salary)  
        FROM Emp  
    )  
);
```

6. Find employees whose salary is greater than the company average salary

```
SELECT Name,  
       Salary  
FROM Emp  
WHERE Salary >  
(  
    SELECT AVG(Salary)  
    FROM Emp  
);
```

7. Find employees whose salary is greater than the average salary of their department

```
SELECT Name,  
       Dept,  
       Salary  
FROM Emp e  
WHERE Salary >  
(  
    SELECT AVG(Salary)  
    FROM Emp  
    WHERE Dept = e.Dept  
);
```

8. Find the minimum salary in each department

```
SELECT Dept,  
       MIN(Salary) AS MinimumSalary  
FROM Emp  
GROUP BY Dept;
```

9. Find the department having the highest average salary

SQL Server

```
SELECT TOP 1  
       Dept,  
       AVG(Salary) AS AvgSalary  
FROM Emp  
GROUP BY Dept  
ORDER BY AvgSalary DESC;
```

MySQL

```
SELECT Dept,  
       AVG(Salary) AS AvgSalary  
FROM Emp  
GROUP BY Dept  
ORDER BY AvgSalary DESC  
LIMIT 1;
```

10. Count employees in each department

```
SELECT Dept,  
        COUNT(*) AS EmployeeCount  
FROM Emp  
GROUP BY Dept;
```

11. Find departments having more than 3 employees

```
SELECT Dept,  
        COUNT(*) AS EmployeeCount  
FROM Emp  
GROUP BY Dept  
HAVING COUNT(*) > 3;
```

12. Find duplicate email addresses

```
SELECT Email,  
        COUNT(*) AS EmailCount  
FROM Emp  
GROUP BY Email  
HAVING COUNT(*) > 1;
```

13. Find the top 3 highest-paid employees

SQL Server

```
SELECT TOP 3 *  
FROM Emp  
ORDER BY Salary DESC;
```

MySQL

```
SELECT *  
FROM Emp  
ORDER BY Salary DESC  
LIMIT 3;
```

14. Display employee names along with department names (Only matching records)

```
SELECT e.Name,  
       d.DeptName  
FROM Employee e  
INNER JOIN Department d  
ON e.DeptID = d.DeptID;
```

15. Display ALL employees along with their department names

```
SELECT e.Name,  
       d.DeptName  
FROM Employee e  
LEFT JOIN Department d  
ON e.DeptID = d.DeptID;
```

16. Display ALL departments along with employees

```
SELECT d.DeptName,  
       e.Name  
FROM Department d  
LEFT JOIN Employee e  
ON d.DeptID = e.DeptID;
```

(Equivalent to a RIGHT JOIN if Employee is written first.)

17. Display all employees and all departments

```
SELECT e.Name,  
       d.DeptName  
FROM Employee e  
FULL OUTER JOIN Department d  
ON e.DeptID = d.DeptID;
```

18. Display each employee along with their manager

```
SELECT e.Name AS EmployeeName,  
       m.Name AS ManagerName  
FROM Employee e  
JOIN Employee m  
ON e.ManagerID = m.EmpID;
```

19. Find customers who have never placed an order

```
SELECT c.CustomerID,  
       c.CustomerName  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerID = o.CustomerID  
WHERE o.OrderID IS NULL;
```

20. Find departments where no employee works

```
SELECT d.DeptID,  
       d.DeptName  
FROM Department d  
LEFT JOIN Employee e  
ON d.DeptID = e.DeptID  
WHERE e.EmpID IS NULL;
```

21. Find products that have never been ordered

```
SELECT p.ProductID,  
       p.ProductName  
FROM Products p  
LEFT JOIN OrderDetails od  
ON p.ProductID = od.ProductID  
WHERE od.ProductID IS NULL;
```

22. Find employees whose Department ID is NULL

```
SELECT *  
FROM Employee  
WHERE DeptID IS NULL;
```

23. Find employees whose Department ID has no matching department

```
SELECT e.Name,  
       e.DeptID  
FROM Employee e  
LEFT JOIN Department d  
ON e.DeptID = d.DeptID  
WHERE d.DeptID IS NULL;
```

24. Combine employee names from two tables (Unique values)

```
SELECT Name  
FROM CurrentEmployees  
  
UNION  
  
SELECT Name  
FROM FormerEmployees;
```

25. Combine employee names from two tables (Including duplicates)

```
SELECT Name  
FROM CurrentEmployees  
  
UNION ALL  
  
SELECT Name  
FROM FormerEmployees;
```